

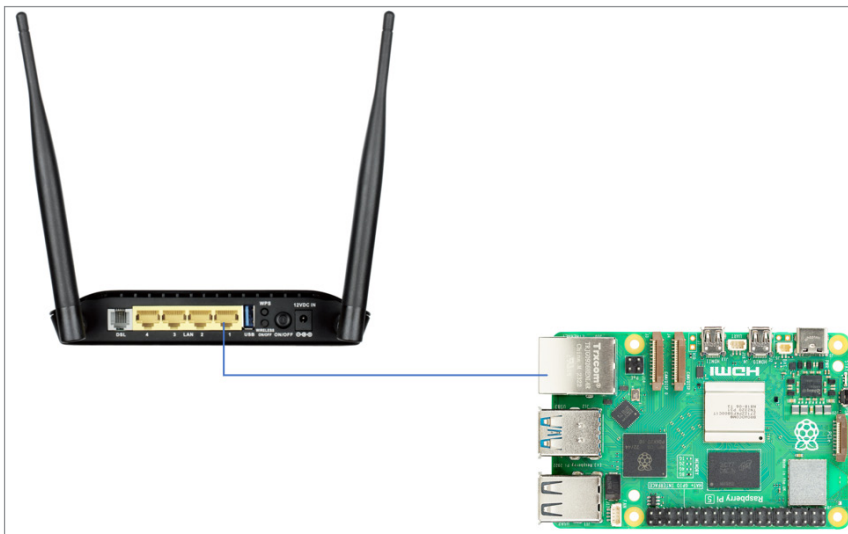


Fig. 3: Wired connection

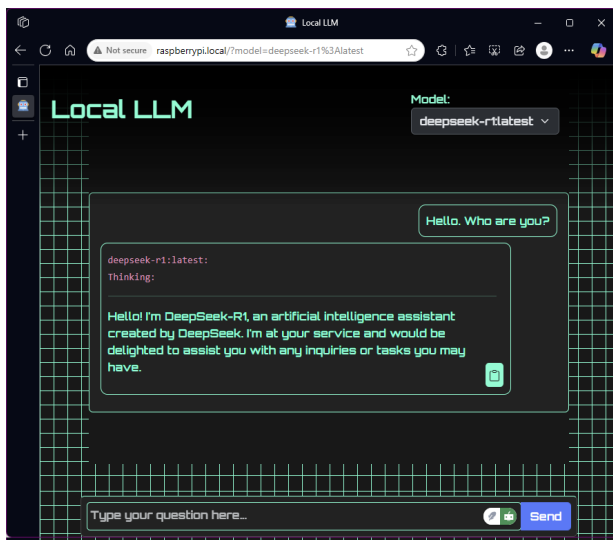


Fig. 4: Using your private LLM

is faster than the standard Raspberry Pi OS.

In the customisation option, select the following:

- Enable the 'Set hostname' field. The default value is 'raspberrypi.local' and need not be changed.
- Enable SSH with password

start flashing. When complete, the SD card can be ejected and put in the Raspberry Pi and powered on. To connect it via Ethernet, you may plug it into the LAN port of the router.

Connect the PC to the home Wi-Fi and SSH to the Raspberry Pi. To do this you may

authentication.

- Set username and password. The default username is 'pi', while the default password is 'raspberry'. It is recommended to not use the default password.

- Set up the Wi-Fi credentials if you are using home Wi-Fi. Otherwise, you can simply connect the device to the router via an Ethernet cable.

You can now

open the command prompt and enter 'ssh pi@raspberrypi.local'. (In a broader sense, 'ssh <username>@<hostname>'.) It will prompt you for the password.

To install Ollama and set it up as home assistant, you may run the following command:

```
curl https://github.com/AdityaMitra5102/LocalLLM/raw/refs/heads/main/oneclickinstall.sh | sh
```

This script automatically installs Ollama and a web UI for the same. It adds environment variables to allow the devices on your network to access it. It also automatically downloads the open source models DeepSeek R1, Llama 3, and Gemma, and runs them privately so that there is no risk of your data leaving your home network.

You can access it at <http://raspberrypi.local> (in a broader sense <http://<hostname>>) from any device connected to your home network.

Locally hosting an LLM comes with its pros and cons. The pros include privacy and data sovereignty, as it gives users complete autonomy over their data and how it is processed. Lower processing power is a disadvantage, and may result in slower outputs as well as an inability to run bigger models. However, as the pros outweigh the cons, LLMs can be locally hosted for most purposes. **EFY**

This article was first published in April 2025 issue of Open Source For You magazine.

Aditya Mitra is a cybersecurity researcher. He likes to learn about the vulnerabilities of various kinds of networks. His areas of interest are IoT, networking, and cybersecurity.

Errata

A defect has been identified in the circuit and PCB published in the article "Automatic Evening Light With Cut-Off Timer" in EFY June 2025 issue. Readers are advised to apply the following correction: In the published design, pin 1 of IC1A is incorrectly connected directly to resistors R5 and R6. These resistors should instead be connected to the anode of LED1. This change ensures that the voltage across capacitor C1 remains within the forward voltage of LED1 (approximately 2V). We regret the oversight and recommend implementing this correction for proper circuit functionality.